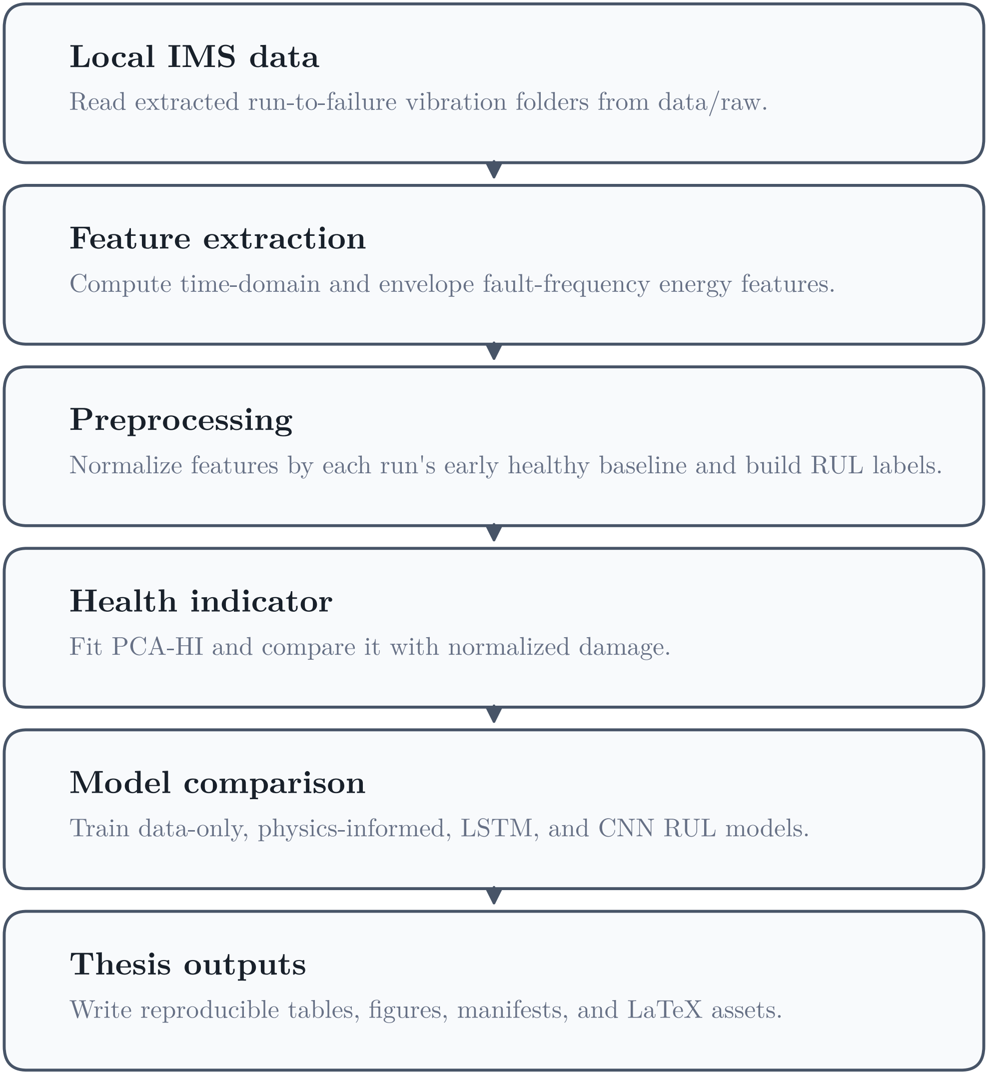
Local IMS data
Read extracted run-to-failure vibration folders from data/raw.
Feature extraction
Compute time-domain and envelope fault-frequency energy features.
Preprocessing
Normalize features by each run's early healthy baseline and build RUL labels.
Health indicator
Fit PCA-HI and compare it with normalized damage.
Model comparison
Train data-only, physics-informed, LSTM, and CNN RUL models.
Thesis outputs
Write reproducible tables, figures, manifests, and LaTeX assets.